AP CHEMISTRY Syllabus 2026-2027
Teacher- Boppana; Room 247
Office Hours : First 20mins of Lunch M-F
Contact: jodi_boppana@mcpsmd.org

Daily Expectations

- **Punctuality & Readiness:** Arrive on time and fully prepared to learn.
- **Personal Mobile Devices (PMDs):** PMDs must be stored out of sight at all times during the school day.
- during the school day.

Grading & Reporting

- All assignment grades will be posted directly in **Canvas and synced to Synergy**
- Students are responsible for regularly monitoring their grades and promptly notifying Ms. Boppana if any discrepancies arise.

Reassessment Policy

- Students may reassess up to **two Assessment Tasks (ATs)** per quarter. PLC Designated
- *Note: Major Unit Assessments are not reassessable.*

Absences & Make-Up Work

- **Notification:** Please send an email to Ms. Boppana as soon as possible when you are absent.
- **Timeline:** Students will have the number of days missed plus one (not to exceed 5) to make up work. Students should arrange a make-up plan with Ms. Boppana right away—the sooner, the better!

Academic Integrity & AI Policy

- All submitted work must be strictly student-owned and adhere to the **Churchill Code of Conduct and Student Expectations**. Unapproved AI usage or plagiarism will result in disciplinary and academic review according to school policy.

Required Daily Materials

- Fully charged, school-issued **Chromebook**
- Writing utensil (**pen or pencil**)
- **AP Reference Packet**

Semester A

Quarter 1

**Unit 1 – Foundations (Part I)

Aug 25 - Sep 14

Read Chapters 1 (include Chapter 4 section 4.3), 2 and 7

Non-instructional days: Sep 7 (Labor Day)

Chapter 1 – Matter and Measurement: An Atomic Perspective

1.1 Exploring the Particulate Nature of Matter
1.3 Classes and Properties of Matter classification & properties of matter, separating mixtures
1.4 States of matter
***1.5** Formulas and models (Go to Chap 4 and read section 4.3: Naming Compounds and Writing Formulas; know the 7 strong acids and 8 strong bases from class discussion)
1.6 Forms of energy Note: Review Coulomb's Law from class discussion, do not use the Potential Energy equation provided to you on page 18 of the textbook. After reviewing Coulomb's Law, be able to explain acid strength in the context of this Law.
1.7 Expressing Experimental Results (review, this is not testable.) The following sections in 1.7 are testable: significant figures, and significant figure calculations. Know these units and their symbols in Table 1.1: kilo, deka, deci, centi, milli, and micro – Note: Metric prefixes are found in your AP Reference packet .)
1.8 Unit conversion and dimensional analysis (Dimensional analysis is how you're expected to do problems in AP Chemistry. If you're not familiar with how to do this, it behooves you to become familiar with the mechanics of this as you'll be graded on whether you do your problems in accordance with dimensional analysis.)

Read the Summary on page 38
Review the Problem-Solving Summary on page 39

Chapter 2 – Atoms, Ions and Molecules: The Building Blocks of Matter

2.1 When Projectiles Bounced Off Tissue Paper: The Rutherford Model of Atomic Structure (Be familiar with all of the experiments of significance around this time, Thomson's cathode ray experiments and his plum pudding model of the atom, Millikan's oil drop experiment, and Bohr's Gold Foil experiment. Understand what alpha and beta particles are.)

2.2 Nuclides and Their Symbols
2.3 Navigating the Periodic Table (The Periodic Table has its own unit in the second semester, take from this section the basics of the PT, specifically where to find metals, nonmetals and semi-metals, and know what charges are associated with what group numbers, specifically for the Representative elements in groups 1, 2, 13 to 17, and know that transition metals have multiple oxidation states (as do most of the nonmetals.)
2.4 The Masses of Atoms, Ions, and Molecules (Know how to calculate the average atomic mass of an element.)
2.5 Moles and Molar Masses (Use 6.02 X 10^{23} for the value of the Avogadro Constant for one mole of anything.)
2.6 Mass Spectrometry: Identifying Molecular Compounds and Determining Their Molecular Masses (Understand how mass spec works.)

Read the Summary on page 76
Review the Problem-Solving Summary on page 77

Chapter 7 – Stoichiometry: Mass Relationships and Chemical Reactions

7.1 Chemical Reactions and the Carbon Cycle (There's nothing specifically testable in this section, it's more FYI.)
7.2 Writing Balanced Chemical Equations (In the past you were expected to be able to "predict" the resultant products of given reactants, that was eliminated from the AP curriculum. You do need to be able to balance chemical equations. You should be able to balance combustion reactions without going through all of the steps outlined here.)
7.3 Stoichiometric Calculations (This is important, and you need to use dimensional analysis for these calculations.)
7.4 Limiting Reactants and Percent Yield (Important, and a recurring topic in many FRQs and MCQs.)
7.5 Percent Composition and Empirical Formulas
7.6 Comparing Empirical and Molecular Formulas (Important and know that ionic compounds do not have "molecular" formulas.)
7.7 Combustion Analysis (This is huge in AP FRQs.)

Read sections 9.6 & 9.7 on the ideal gas law and its use in determining density of a gas

Read the Summary on page 327 - 328
Review the Problem-Solving Summary on page 328

****Tuesday 9/15: AP Chem Exam #1 – Stoichiometry (This exam covers Chaps 1, 2 and 7)**

**Unit 2 – Foundations (Part II)
Sep 16 – Sep 27

Non-instructional Days: Sep 21 (Yom Kippur); Early release days: Fri Sep 18

Read Chapter 8

Chapter 8: Reactions In Aqueous Solutions

- **8.1 *Solutions and Their Concentrations*** (A very important section. We are primarily concerned with Molarity which is the primary unit of concentration used in chemistry – there are other concentration units, but Molarity is what we'll be using in AP chemistry.)

- ***8.2 *Dilutions*** (Very important, know how to create a solution and how to use a volumetric flask as discussed here**. In the section that deals with "Determining Concentration", review this, it will be addressed again in the 2nd semester and it's not testable for this unit exam**.)

- ***8.3 *Electrolytes and Nonelectrolytes*** (This is focused on terminology, it's important that you know this, especially when we head into acids and bases where most of what we're dealing with are weak acids and bases.)

- ***8.4 *Acid–Base Reactions***: Proton Transfer (There's information here that's not going to be on the unit test on 9/28 but will be on the unit test for acids and bases. What you need to know starts on page 352 and this addresses net-ionic equations, and this is important.)
- ***8.5 *Titrations*** (Quick review of this unit, it's not testable on this unit test but will be for the acids and bases unit test.)

- **8.6 *Precipitation Reactions*** (Know which ions **don't** form precipitates and understand what it means that a precipitate forms, but you do not need to know specific precipitation reactions.)

- **8.7 *Oxidation–Reduction Reactions:*** Electron Transfer (A large part of this unit's exam, you need to understand reduction and oxidation and how to balance ReDox Reactions using the half-reaction method.)

Read the Summary on page 387
Review the Problem-Solving Summary on page 388

Monday 9/28: AP Chem Exam #2 – Reactions (This exam covers Chapter 8)

**Unit 3 – Chemical Equilibrium
9/29 – 10/8

Read Chapter 14

Chapter 14: Chemical Equilibrium: Equal but Opposite Reaction Rates
9/29 – 10/8

***14.1** The Dynamics of Chemical Equilibrium (Read this section knowing that there are parts of it that we haven't covered yet, specifically Kinetics and Thermodynamics of chemical reactions, which will be covered later.)

14.2 Writing Equilibrium Constant Expressions

14.3 Heterogeneous Equilibria

14.4 Manipulating Equilibrium Constant Expressions

***14.5** Equilibrium Constants and Reaction Quotients (Q, the reaction quotient, is essential for understanding where an equilibrium reaction is at and where it will go.)

14.6 Le Châtelier's Principal

14.7 Calculations based on K

***14.8** Equilibrium and Thermodynamics (Similar to 14.1, this section covers thermodynamics and will be revisited in the next two units. Read this but you won't be tested on the thermodynamic aspects addressed here.)

***14.9** Changing K with Changing Temperature Know that the effect of temperature on an equilibrium reaction is directly tied to whether the. reaction is exothermic or endothermic, and be able to determine what direction the equilibrium will shift based on that.)

Read the Summary on page 704
Review the Problem-Solving Summary on page 705

Friday 10/9: AP Chem Exam #3 – Equilibrium (This exam covers chapter 14.)

This is the last summative assessment for the first quarter, which ends on 10/30

First quarter ends 10/30, second quarter begins 11/4

**Unit 4: Acid – Base Equilibria
10/10 – 11/6

Non-instructional Days: 10/16 Professional Day, no classes; 11/2 Grading and planning for teachers, no classes; 11/3 Election Day

Read Chapter 15,

Chapter 15: Acid – Base Equilibria – Proton Transfer in Biological Systems

Memorize: strong acids and Strong Bases

15.1 Acids and Bases: A Balancing Act
15.2 Ka, Kb, and the Strengths of Acids and Bases
15.3 Conjugate Pairs
15.4 pH and the Autoionization of Water
15.5 Ka, Kb, and the Degree of Ionization of Acids and Bases
15.6 Calculating the pH of Acidic and Basic Solutions
15.7 Polyprotic Acids
15.8 Acidic and Basic Salts

Read the Summary on page 755
Review the Problem-Solving Summary on page 756

Chapter 16: Additional Aqueous Equilibria: Chemistry and the Oceans

16.1 Ocean Acidification: Equilibrium under Stress
16.2 The Common-Ion Effect
16.3 pH Buffers
16.4 pH Indicators and Acid–Base Titrations
16.5 Lewis Acids and Bases (FYI, this is not testable)
16.6 Formation of Complex Ions (Understand complex ion formation, you don't have to name them or understand much beyond that they form, and this helps to explain section 16.7.)
16.7 Hydrated Metal Ions as Acids (Know that some metals, in particular those with a +2 or greater charge, will act as acids or bases in solution and WILL affect pH.)

Read the Summary on page 813
Review the Problem-Solving Summary on page 813

Monday 11/9: AP Chem Exam #4 – Acids and Bases

Unit 5: Thermodynamics
11/10 – 12/02

Non-instructional Days: 11/26 – 11/27 (Thanksgiving); Early release days: 11/23, 11/24, and 11/25

Read Chapters 10 and 11

Chapter 10 Thermodynamics — Part I: Enthalpy and the Transfer of Energy in Chemical Systems

10.1 Energy as a Reactant or Product
10.2 Transferring Energy and Doing Work
10.3 Enthalpy and Enthalpy Changes
10.4 Heating Curves and Heat Capacity
10.5 Enthalpies of Reaction and ***Calorimetry***
10.6 Hess's Law and Standard Enthalpies of Reaction
10.7 Enthalpies of Reaction from Enthalpies of Formation and Bond Energies
10.8 Enthalpy Changes When Substances Dissolve
10.9 More Applications of ***Thermochemistry***

Read the Summary on page 501
Review the Problem-Solving Summary on page 502

Chapter 11 Thermodynamics—Part II: Entropy, Gibbs Energy, and Why Some Processes Happen and Others Do Not

11.1 Why Chemical Reactions Happen
11.2 Entropy and the Second Law of Thermodynamics
11.3 Absolute Entropy and Molecular Structure
11.4 Applications of the Second Law
11.5 Calculating Entropy Changes
11.6 Gibbs Energy
11.7 Dependence of Thermodynamic Favorability on Temperature
11.8 Driving the Human Engine: Coupled Reactions

Read the Summary on page 546
Review the Problem-Solving Summary on page 547

Thursday 12/3: AP Chem Exam # 5: Thermodynamics Parts I & II

Unit 6: Electrochemistry

12/4 – 12/18

Non-instructional Days: 12/24 – 1/4 (Winter break); Early release days: 12/23

Read Chapter 17: Electrochemistry:

17.1 Running on Electrons: Redox Chemistry Revisited
17.2 Electrochemical Cells
17.3 Electrode Potentials
17.4 Cell Potentials E_{cell}
17.5 Voltaic Cells at Work
17.6 The Effect of Concentration on E_{cell}
17.7 Relating Battery Capacity to Quantities of Reactants
17.8 Corrosion: Unwanted Electrochemical Reactions
17.9 Electrolytic Cells and Rechargeable Batteries
17.10 Fuel Cells

Read the Summary on page 859; Review the Problem-Solving Summary on page 860

Monday 12/21: AP Chem Exam #6: Electrochemistry

Unit 7: Chemical Kinetics

1/4 – 1/15

Non-instructional Days: 1/18 MLK Day

Read Chapter 13 Chemical Kinetics: Improving Air Quality

13.1 Air Quality
13.2 Reaction Rates
13.3 Effect of Concentration on Reaction Rate
13.4 Reaction Rates, Temperature, and the Arrhenius Equation
13.5 Reaction Mechanisms
13.6 Catalysts
Read the Summary on page 647
Review the Problem-Solving Summary on page 648

***16.8 Solubility Equilibria

Read the Summary on page 813
Review the Problem-Solving Summary on page 813

Tuesday 1/19: AP Chem Exam #7: Chemical Kinetics and Solubility Equilibria

The second quarter ends 1/22., and the third quarter and second semester begin on 1/26